

Rose number

Rose number(Four-leaf Rose Number)[四葉玫瑰数]

Rose number is a Narcissistic number (自幂数) or Armstrong number (阿姆斯壯数)

This refers to a 4-digit number where the sum of the fourth powers of its digits equals the number itself.

Example: 1634

$$1^4 + 6^4 + 3^4 + 4^4 = 1634$$

So 1634 is a Four-leaf Rose Number.

There have 3 rose number is:

1. 1634
2. 8208
3. 9474

So it have a problem that is when we forget those number, can we find that?
Yes, I some of the method to find those number.

1 Find Four-leaf Rose Number with programming.

1.1 In C++

```
1      #include<iostream>
2      #include<cstdio>
3      using namespace std;
4
5      int main(){
6          int a,b,c,d;
7          for (int num=1000;num<=9999;num++){
8              a = num / 1000;
9              b = (num / 100) % 10;
10             c = (num / 10) % 10;
11             d = num % 10;
12             if (a*a*a*a+b*b*b*b+c*c*c*c+d*d*d*d==num)
13                 printf("%d\n",num);
14         }
15         cin.get();
16         return 0;
17     }
```

1.2 In python

```
1      for num in range(1000,10000):
2          a = num // 1000
3          b = (num // 100) % 10
4          c = (num // 10) % 10
5          d = num % 10
6          if a**4+b**4+c**4+d**4==num:
7              print (num)
```